

Chemistry Paper 4 Notes

Precision

Instrument	Uncertainty in measurement	Uncertainty in calculation
<u>Volume</u>		
Pipet	0.04/0.06 cm^3	1 <i>d.p.</i>
Burette	0.1 cm^3	2 <i>d.p.</i>
100 cm^3 Measuring cylinder	0.5 cm^3	3 <i>d.p.</i>
250 cm^3 Measuring cylinder	1 cm^3	
<u>Mass</u>		
Electronic Mass Balance	0.1 <i>g</i>	
Spring Balance	5 <i>g</i>	
<u>Temperature</u>		
Mercury thermometer	0.1 °C	
Alcohol thermometer	0.5 °C	
<u>Time</u>		
Stopwatch	<i>Reaction time (0.2s)</i>	

General Experimental Techniques

1. Weighing mass

- Subtraction method

Mass of *container* /g

Mass of *container and chemical* /g

Mass of *container and residue chemical* /g

Mass of *chemical* used /g

- Tare method

Mass of chemical used = x g (TARE)

2. Titration

Titration	1	2
Initial burette reading / cm ³		
Final burette reading / cm ³		
Volume of solution used / cm ³		

$$\text{Average volume of solution used} = \frac{V_1 + V_2}{2}$$

***Repeat to obtain two consistent titres within 0.10 cm³ in difference**

***Add titrant from burette dropwise near end point**

- Types of titration:
 - Acid-base
 - WRITE OUT EQUATION (pay attention to coefficients when diprotic acid/base is used)
 - Redox
 - Thermometric
- Indicator (for acid-base titration)

Titration	pH change	Indicator	Colour change
strong acid – strong base	4 – 10	methyl orange (3.1 – 4.4)	red – orange – yellow
		thymolphthalein (9.3 – 10.5)	colourless – pale blue - blue
strong acid – weak base	3.5 – 6.5	methyl orange (3.1 – 4.4)	red – orange – yellow
weak acid – strong base	7.5 – 10.5	thymolphthalein (9.3 – 10.5)	colourless – pale blue - blue
weak acid – weak base	no marked pH change	no suitable indicator (a pH meter could be used and end-point can be determined by plotting a graph of pH against volume)	

- Indicator is suitable if:
 - working range of indicator falls within rapid pH change
 - colour change is distinct
- $\text{Change in concentration} = \frac{\text{Change in amount}}{\text{Volume}}$
- If a less concentrated titre is used, titration will be more accurate as for the same absolute uncertainty, a higher volume of titre is used, and hence smaller percentage error.
- If burette tip is not filled, the titre value will be higher than expected. The calculated amount of ... will be higher than expected. The calculated amount of the ... will be smaller than expected. Hence, the calculated percentage purity of the sample will be lower than actual.
- If apparatus (pipette) is not rinsed with solution, deionised water in the apparatus dilutes the ... solution. Since less ... is transferred into the conical flask, less ... will be produced for titration. Therefore, ... will be required for complete reaction.
Titre obtained will be lower than expected.
- Water of crystallization is usually less than 10
- Percentage uncertainty has a $\pm$!!!
- Gravimetric method is less reliable than titrimetric method as:
 - Temperature of dehydration cannot be controlled using a Bunsen burner. Hence, it is difficult to prevent decomposition of anhydrous salt when it is first formed.
Improvement: Carry out the dehydration in a vacuum oven under controlled temperature of 400 °C.
 - Anhydrous salt may absorb moisture from the air during cooling.
Improvement: Cool anhydrous salt in a desiccator before weighing.

Graph

- Extrapolate lines until “meet” vs “cross”

Energetics

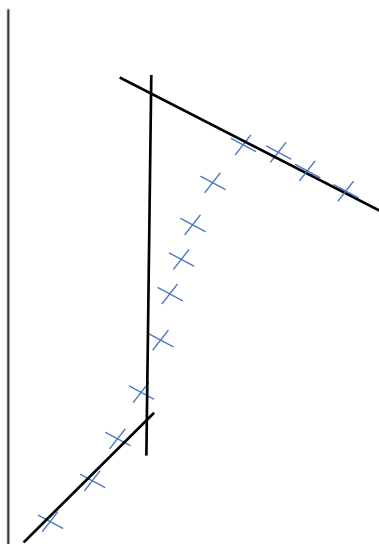
Aim: Find ΔH or $[FAX]$ or percentage by mass by measuring ΔT ($\Delta T = T_f - T_i$, $T_i =$ **weighted average** $\frac{T_1 V_1 + T_2 V_2}{V_1 + V_2}$).

***Remember negative sign!!!**

A. Calorimetry

- Reactions with two solutions
 - e.g. Neutralization
 - $\Delta H = \frac{Q}{n} = \frac{mc\Delta T}{n}$, where n is the mole of water
- Reactions with solid and solution
 - e.g. metal + acid
 - $\Delta H = \frac{Q}{n} = \frac{mc\Delta T}{n}$, where n is the mole of limiting agent

Plot cooling curve

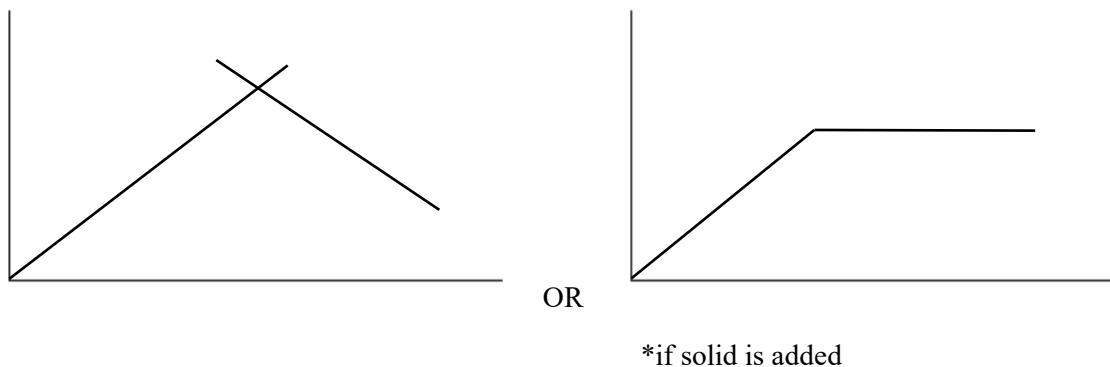


* Extrapolate to account for heat loss to environment

B. Thermometric Titration

Volume of x OR Mass of x	Highest temperature

Plot two best-fit-line graph



- If a cooler environment is used, the titre value will be higher than expected. The lower temperature of environment results in the lower rate of reaction \
- Weak bases give a lower T_{max} and hence a lower ΔT .
Ethanoic acid, a weak acid, is partially dissociated in water whereas hydrochloric acid, a strong acid, is fully dissociated. Some of the heat evolved from neutralization is absorbed to dissociate the ethanoic acid molecules completely, resulting in a smaller ΔT at the equivalence point.
- Using a diprotic acid with half the concentration of the monoprotic acid used has no change on the mass of solid needed as same amount of H^+ is involved.
- Initial temperature of all solutions needs to be accounted for by using weighted initial temperature.
- Crucible is preferred over a boiling tube as it has a larger opening which allow the water vapour formed to escape easily.
- ΔH is different from ΔH calculated using bond energy values.
 - ΔH assumes that all substances are in their standard states. However, X is in liquid / solid state while bond energies are calculated when it is in gaseous state.
 - Intermediate bond character: single bond having double bond character / double bond having triple bond character due to the delocalization of π electron over molecule
 - Bond strain: Weakens the bond. Less energy is required to break the bond.

Sources of error	Precautions
Heat was lost to the surroundings. The maximum temperature recorded is lower than actual, resulting in a lower temperature rise than actual. Thus, the calculated heat evolved is lower than actual and ΔH is less negative than actual.	Carry out the experiment in a <u>draught-free environment</u> to minimise heat loss to the surroundings. Use a <u>polystyrene cup with a lid on</u> to carry out the reaction.
The 50 cm^3 measuring cylinder has an absolute uncertainty of 0.5 cm^3 . This resulted in <u>large percentage uncertainty</u> for the volume of	<u>Use a burette instead</u> of a measuring cylinder give rise to lower absolute uncertainty of 0.10 cm^3 for volume of solution measured.

solutions measured. Thus, the <u>calculated heat evolved and ΔH are inaccurate.</u>	Each burette reading has an absolute uncertainty of 0.05 cm^3. Due to lower percentage uncertainty for measured volumes of the solutions, the results are more accurate.
The <u>thermometer reading has an absolute uncertainty of 0.1°C</u> and the measured temperature rise is very small. This resulted in <u>large percentage uncertainty</u> for the temperature rise. Thus, the calculated heat evolved and ΔH are inaccurate.	<u>Use a digital thermometer</u> with a smaller absolute uncertainty to measure temperature more accurately.
There is a <u>loss of material</u> if the solution overflows or sprays out.	<u>Use a larger cup/ container.</u> Cover the cup with a lid to minimise spray.

Reaction kinetics

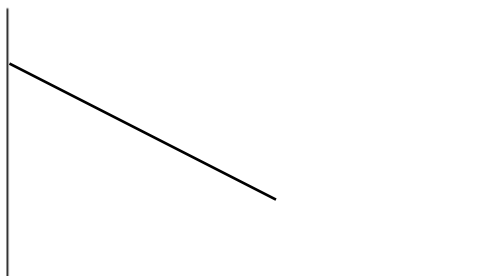
Aim: Find order of reaction by measuring volume (since $volume \propto concentration$ when total volume is constant) or rate by measuring $\frac{1}{time}$ AND

plotting graph of **$\lg Rate = n \lg[x] + \lg k$** , where n is the order of reaction.

A. Titrimetric method – Continuous *Used when there is no visible change*

- Samples are collected from the reacting mixture and quenched by:
 - Rapid cooling by plunging mixture into ice
 - Diluting by adding a large quantity of water
 - Adding a quenching reagent which stops/slows down the reaction by completely removing a reagent
- Samples are then titrated to determine amount of reactant / product at each sampling time.

Change in conditions	Result
If <u>a cooler environment is used</u>	Titre value will be higher than expected. The lower temperature of environment results in the lower rate of reaction and hence a larger amount of product/reactant remains present in sample. A larger volume of x is required for titration. • A water / oil bath can be used to maintain the temperature at room temperature for the reaction.
If <u>a much higher concentration of reactant x is used</u>	It remains relatively constant as very little proportion of it is consumed for reaction. Reaction is pseudo first order Hence, rate with respect to the specific reactant can be studied.
If experiment is <u>done on another day</u>	Temperature have changed OR H_2O_2 would have decomposed



Zero order:



B. Clock reaction – Discontinuous *Used when there is a visible change*

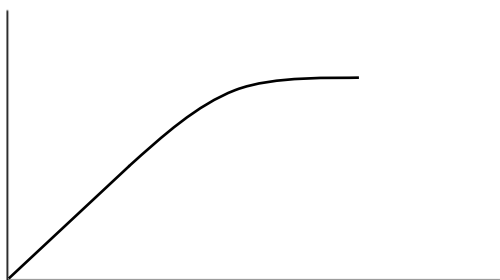
- Measure time for visible change to occur
- Keep total volume constant such that volume can be used to represent concentration.
- Source of error:
 - Difficult to determine when visible change occurs
 - Human reaction time

Change in conditions	Result
Using a 250 cm^3 instead of a 100 cm^3 beaker	Results in a lower height of solution. More product is required to obstruct the same amount of light. Hence, <u>time taken would be longer.</u>
Stopwatch is started when about half of the solution has been added.	If stopwatch is only started when all the solution is added, the <u>reaction would have started before the stopwatch is started.</u> This will result in the <u>experimental timing being shorter than actual.</u>

- Mn^{2+} ions act as a catalyst by providing an alternative pathway of lower activation energy. The reaction between MnO_4^- and $\text{C}_2\text{O}_4^{2-}$ involves two anions that repel each other, thus requires a high activation energy.
- $\frac{1}{t}$ can be used as rate as they are proportional only when total volume is fixed
If volume used in each experiment is different, rate is no longer proportional to $\frac{1}{t}$.

C. Gravimetric / Mass loss method (Gas collection method)

- Volume of gas produced per unit time
- Mass loss per unit time



- To collect a dry sample of gas, use a well-greased gas syringe instead of downward displacement of gas method.
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Sources of error	Precautions
The thermometer reading has an absolute uncertainty of $0.1\text{ }^\circ\text{C}$ and the measured temperature rise is very small. This resulted in large percentage uncertainty for the temperature rise. Thus, the calculated heat evolved and ΔH are inaccurate.	Use a digital thermometer with a smaller absolute uncertainty to measure temperature more accurately.

The volume of gas collected by downward displacement of gas is less than actual volume as some gas (CO_2) might have dissolved in water. Thus, amount of gas (CO_2) calculated is less than actual and the percentage of $CaCO_3$ in sample solid is less than expected.	Use a well-greased gas syringe.
During cooling process metal oxide may react with the air to form hydroxides or carbonates.	Cool in <u>desiccator</u>
To evaporate water of crystallization without decomposing the metal	Carry out the dehydration process in a <u>vacuum oven</u> under controlled temperature.

Planning

- Quantity
- Apparatus
- Specific action

Method

- State choice of apparatus
- Procedures
 - Weighing solids
 1. Place **clean and dry container** on the weighing pan and close the draft shield doors.
 2. Weight mass of container accurately.
 3. Add required mass of solid into container and record the mass of container and its content.
 4. Calculate mass of solid transferred using subtraction method.
 - Preparing solution by dilution
 1. Transfer solid into **CLEAN and DRY beaker**.
 2. Add small volume of deionised water to completely dissolve the solution.
 3. Transfer solution to a volumetric flask.
 4. Top up volumetric flask with deionised water until graduation mark.
 5. **Stopper the volumetric flask** and **shake to ensure homogeneous solution**.
 - Preparing saturated salt solution
 1. Transfer weighted solid into a **CLEAN and DRY** beaker
 2. Add small volume of deionised water/specific solvent into the beaker and **stir gentle** using a glass rod to dissolve as much of solid as possible.
 3. Repeat step 2 **until some solid remains undissolved** to **ensure a saturated solution is obtained**.
 4. **Leave the mixture to stand in constant temperature water bath** for at least 30 minutes to **allow equilibrium to be reached**.
 5. Filter the saturated solution into a **DRY conical flask** using a **DRY filter funnel and filter paper** to **ensure the saturated solution is not diluted**.
 - Heating solid (thermal decomposition)
 1. Weigh mass of **EMPTY, CLEAN and DRY crucible** / boiling tube.
 2. Add solid and weigh and record mass of *container + solid*.
 3. Heat solid in boiling tube at 45° using non-luminous flame for at least 10 minutes.
 4. Cool to room temperature and weigh mass of *container + its content*.

5. Repeat step 3 and 4 until mass of *container + its content* is **within 0.050 g in difference.**

○ Collecting gas

<u>Insoluble / Sparingly soluble</u>	<u>Moderate</u>	<u>Very soluble</u>
CO, CO_2, H_2, N_2, O_2	SO_2, Cl_2	NH_3, HCl, HBr, HI

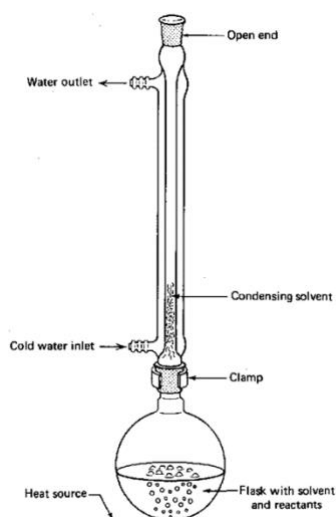
Insoluble gas – Downward displacement of water

Set up:

Soluble gas – Well-greased Gas syringe

Set up:

○ Reflux set-up



*Open end is to **prevent pressure from building up** and **causing breakage**

Variable / Measurement / InformationEnergeticsThermal decomposition

Mass of porcelain boat /g	A	
Mass of porcelain boat and solid /g	B	
Mass of solid /g	C	
	1 st Heating	2 nd Heating
Mass of porcelain boat and solid /g	D	E

, where $D - E < 0.050 \text{ g}$

Precaution / Reliability / Safety / Accuracy

- Always use **BURETTE**
- Turn off gas before titrating
- Switch off hot place after use
- Use a towel to handle the hot conical flask
- Hold the boiling tube with a test-tube holder to avoid direct contact with the hot boiling tube
- Use mortar and pestle with care
- **DRY** Styrofoam cup
- Set-up must be secured to prevent toppling and breakages
- Secure polystyrene cups by placing it in a beaker to prevent toppling
- Use a towel to hold the beaker
- Carry out the experiment in a **draught-free environment** to minimise heat loss to the surroundings.
- **Use a burette instead of a measuring cylinder** give rise to lower absolute uncertainty of 0.10 cm^3 for volume of solution measured.
- Use a **digital thermometer** with a smaller absolute uncertainty to measure temperature more accurately.
- **DRY** filter paper and funnel and beaker
- **DRY** and **CLEAN** crucible
- **COOL** crucible to **room temperature** before weighing
- **Heat at least 5 minutes** for every heating cycle
- Make sure residue mass is **consistent within 0.050g in difference**
- Use a pair of **metal tongs** to handle the **hot crucible** / **boiling tube**.
- Use **well-greased gas syringe** to collect gas.
- *IF H_2 GAS EVOLVED* Pass an inert gas through the apparatus to flush out the H_2 gas.

Titration

- Specify **indicator** + **colour change**
- Add titrant from burette **dropwise near end point**
- **Repeat** to obtain **two consistent titres within 0.10 cm^3 in difference**

Electrochem

- Place **CLEAN and DRY** beakers in **thermostatically controlled water bath**
- Salt bridge: Soak **strip of filter paper in saturated potassium nitrate solution**